

Element K

Bryson Preddy

- defining the problem (elements A-C) **The problem that we had was that the top was too heavy for the motor we had before, after that we got a bigger motor and a lighter roof. The only problem with the lighter roof was that the program would not work, because just 1 second would make the roof go up and down because the roof is so light, it would basically overlap with the string.**
- ideation (element D) **For the idea we took mostly Jacob's idea and subbed Bryson's in just a little.**
- testing (elements H-I) **We tested if the motor was strong enough to lift the roof and if the roof needed to be lighter. We would know if the roof was to light because the program would not work because the string on the motor would overlap.**

Lessons Learned About the Engineering Design Process (K3)

In my opinion designers might consider making many designs so that they can be tested out in the real world. I would also consider them to think about the weather because we did not really think about the weather and it affected us, but to be honest we didn't really have time to make a weatherproof kind of a project, we finished the project about a week before the actual presentation.

A level 5 reflection illustrates thorough and specific details on improving the design process in the future.

We started off with Jacob's and Bryson's sketch. Then we had to discuss how we were going to combine them into a whole to make the project. We made the project out of wood and we made a roof so that way it would cover up the plant pretty good. We saw that the poles of wood were all level and that wouldn't help the roof lift up, so we cut off an inch to help the proof lift up. The motor that we had to use was not enough volts to lift up the roof so we cut out the inside of the roof but it was still too heavy for the battery. We then got a bigger motor but by the time the motor came in we had built a new roof that was made out of plastic and a lot lighter than the first roof. After all of that we finally finished up the project by adding a rope to the motor so it would lift up the roof by itself.